

Phosphorylation-dependent modulation of presynaptic function/plasticity

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Abstract

Presynaptic plasticity is a fundamental determinant of synaptic efficacy, yet the molecular mechanisms governing its expression remain incompletely understood. This dissertation examines the hypothesis that protein phosphorylation constitutes a unifying regulatory mechanism coordinating presynaptic plasticity across multiple timescales. Through systematic analysis of 82 peer-reviewed studies, phosphoregulation is traced across the synaptic vesicle cycle, from reserve pool mobilisation and active zone priming to Ca^{2+} -triggered fusion and endocytic recovery. The evidence demonstrates that discrete kinase and phosphatase cascades – including PKA, CaMKII, CDK5, PKC, and ERK – act on convergent substrates such as Synapsin I, Munc18-1, RIM1 α , and dynamin I to bidirectionally modulate release probability, vesicle pool size, and active zone architecture. A multi-phosphocode model is proposed, wherein combinatorial phosphorylation states – rather than single phosphosite events – encode short-term, long-term, and homeostatic plasticity.

Declaration

I, Serin Baek, confirm that the work presented in this dissertation is my own. Where information has been derived from other sources, I confirm that this has been indicated in the dissertation.

Acknowledgements

This dissertation would not have been possible without the guidance of my supervisor, Professor Talvinder Sihra, a fountain of knowledge. I thank God for his guidance throughout my studies, Soli Deo Gloria.

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1. Introduction

Synaptic plasticity, the activity-dependent modulation of transmission strength, is a fundamental cellular mechanism underlying learning and memory (Bliss and Collingridge, 1993; Kandel, 2001). Synaptic transmission is stochastic, governed by the release probability (P_r), the number of active release sites (N), and quantal size (Q). Presynaptic plasticity is expressed as changes in P_r and N throughout the synaptic vesicle cycle (Saviane and Silver, 2007). These changes span multiple timescales: short-term plasticity (STP), encompassing paired-pulse facilitation and depression, augmentation, and post-tetanic potentiation (PTP), operates over milliseconds to minutes and reflects residual Ca^{2+} , vesicle availability, and release machinery state (Appendix 1; Katz and Miledi, 1968; Zucker and Regehr, 2002). Long-term potentiation (LTP) and long-term depression (LTD) persist for hours or longer (Deng and Klyachko, 2011), while homeostatic plasticity operates over still longer timescales to maintain stable network output during sustained perturbation (Pozo and Goda, 2010; Delvendahl and Müller, 2019). Although early work focused predominantly on postsynaptic mechanisms, accumulating evidence identifies the presynaptic terminal as a major locus of plasticity across all timescales (Castillo et al., 2002; Chi, Greengard and Ryan, 2003).

1.1. Phosphorylation regulates presynaptic plasticity

Protein phosphorylation – the reversible, ATP-dependent transfer of a phosphate group to serine, threonine, or tyrosine residues – is ideally suited to coordinate presynaptic plasticity (Figure 1; Nestler and Greengard, 1999). It is rapid, spatially constrained, and bidirectional, enabling precise activity-dependent control of protein conformation, interactions, and stability.

Phosphorylation operates across all major forms of presynaptic plasticity: kinase-dependent vesicle mobilisation and priming efficiency underlies STP (Jovanovic et al., 2001), PKA-dependent phosphorylation of RIM1 α is required for mossy fibre LTP (Lonart et al., 2003), and coordinated kinase-phosphatase activity adjusts P_r and vesicle pool size during homeostatic compensation – as evidenced by the widespread, rapid phosphorylation changes triggered by stimulation of isolated nerve terminals (Kohansal-Nodehi et al., 2016).

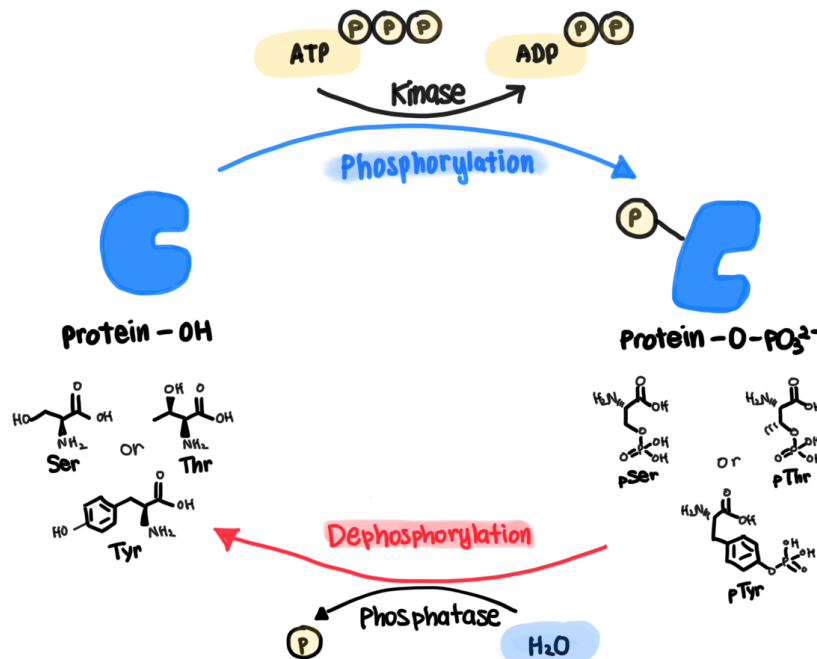


Figure 1. Phosphorylation as a reversible molecular switch regulating presynaptic protein function. Phosphorylation involves the ATP-dependent transfer of a phosphate group to serine, threonine or tyrosine residues on target proteins by protein kinases, whereas protein phosphatases remove phosphates. Phosphate addition introduces negative charge and can alter protein conformation, binding interactions, localisation and activity.

1.2. Thesis and objectives

These observations identify protein phosphorylation as a unifying regulatory mechanism of presynaptic plasticity, yet how distinct pathways selectively modulate different plasticity forms remains unclear. This dissertation tests the hypothesis that specific kinase and phosphatase cascades modulate vesicle availability, active zone function, and Ca²⁺-release coupling to shape presynaptic plasticity. Specifically, I aim to: (i) determine how phosphorylation regulates vesicle pool mobilisation and replenishment during STP; (ii) identify phosphorylation events required for presynaptic LTP at hippocampal mossy fibre-CA3 synapses; and (iii) define how phosphoregulation contributes to homeostatic stabilisation of neurotransmitter release.

2. Methods

2.1. Literature search and data analysis

A structured literature search of PubMed and Web of Science (1985 - 6 January 2026) used the Boolean query ("presynaptic plasticity" AND "phosphorylation"). Inclusion was limited to peer-reviewed studies in English examining defined presynaptic phosphoproteins and associated kinase/phosphatase pathways, primarily at mammalian central synapses. Studies restricted to postsynaptic mechanisms were excluded. From 525 screened records, 82 papers met eligibility criteria after duplicate removal (Figure 2). Studies were then grouped by the functional stage of the synaptic vesicle cycle at which each phosphoregulated protein acts (docking, priming, fusion, endocytosis), enabling systematic comparison of kinase/phosphatase influence across the cycle.

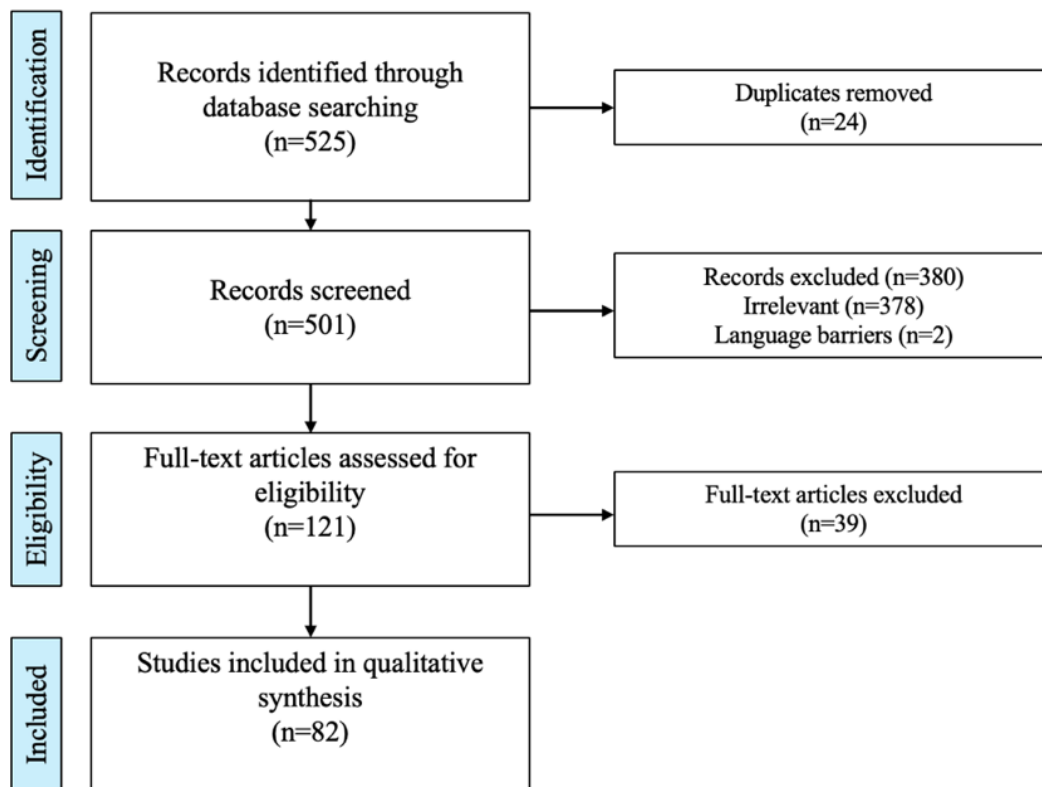


Figure 2. PRISMA Flow Diagram of Search Strategy and Study Selection. Search was conducted according to PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines. Created with Microsoft Word.

3. Protein phosphorylation regulates synaptic vesicle cycling

Neurotransmitter release forms the basis of neural communication (Südhof, 1999). At its core lies the synaptic vesicle cycle – comprising mobilisation, active zone docking, priming into a fusion-competent state, Ca^{2+} -triggered fusion, and endocytic recycling (Appendix 2) – which transcribes presynaptic electrical activity into chemical signals (Alabi and Tsien, 2012).

Throughout this cycle, vesicles are organised into functionally distinct pools – the readily releasable (RRP), recycling, and reserve pools – whose dynamic exchange governs synaptic output (Rizzoli and Betz, 2005; Appendix 3). Phosphorylation modulates this cycle at multiple nodes, regulating vesicle-cytoskeleton interactions, active zone organisation, and the fusion machinery to shape presynaptic plasticity (Chi, Greengard and Ryan, 2003; Zucker and Regehr, 2002).

3.1. Vesicle mobilisation and active zone recruitment: Synapsin I as a convergence point for phosphoregulatory signals

The dynamic recruitment of vesicles from the reserve and recycling pools to the active zone is a primary control point for synaptic strength, governed by a network of phosphoproteins regulated by neuronal activity (Figure 3; Greengard et al., 1993). Central to this network is Synapsin I, the most abundant synaptic vesicle phosphoprotein, which formed liquid condensates to maintain the reserve pool, tethering vesicles to each other and to the actin cytoskeleton (Milovanovic et al., 2018). Action potential-driven Ca^{2+} influx induced the activation of CaMKII and PKA, whose phosphorylation of Synapsin I at sites 1-3 (PKA on site 1 and CaMKII on sites 2 and 3) weakened vesicle-actin interactions and promoted mobilisation from the reserve to the recycling pool (Figure 4; Sihra et al., 1989; Benfenati et al., 1992; Jovanovic et al., 2001; Chi, Greengard and Ryan, 2003). Consistently, dephosphorylation or loss of Synapsin I reduced the recycling pool and evoked release, while calcineurin-dependent dephosphorylation at MAPK-sensitive sites 4-6 limited glutamate release (Llinás et al., 1985; Sihra et al., 1995; Jovanovic et al., 2001). Consequently, Synapsin KO mice exhibited seizures and synapsin II KO showed reduced PTP and accelerated synaptic depression (Rosahl et al., 1995).

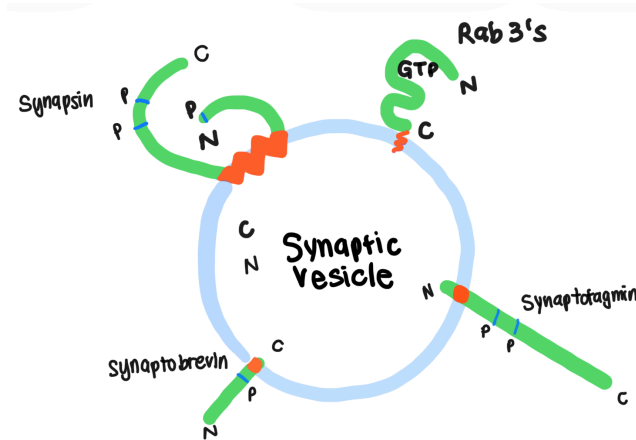


Figure 3. Structures of major synaptic vesicle proteins involved in vesicle trafficking.

Schematic, approximately to scale, of the principal synaptic vesicle proteins implicated in SV trafficking. Isoprenyl groups that anchor Rab proteins are shown as thin orange lines. Protein orientation is indicated by N- and C-termini, and P marks known phosphorylation sites.

Modified from Südhof, 2004.

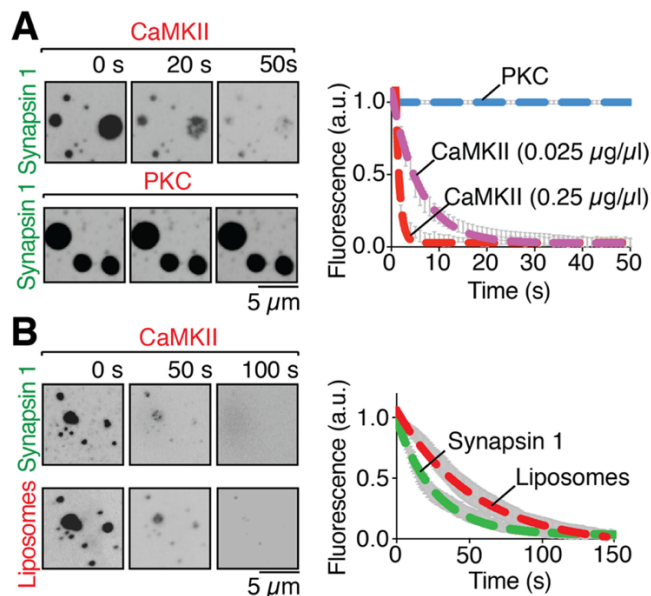


Figure 4. CaMKII-dependent phosphorylation disperses Synapsin I condensates and releases vesicle-associated assemblies. (A) Time-lapse imaging shows that CaMKII, but not PKC, disrupts EGFP-synapsin I condensates in vitro. Right panel quantifies fluorescence loss from individual droplets, reflecting condensate dispersal. (B) Similar dispersal is observed for condensates containing synapsin I and fluorescent liposomes, which mimic synaptic vesicles, indicating coordinated release of vesicle-like structures. Adapted from Milovanovic et al., 2018.

3.1.1. Phosphorylation-dependent control of vesicle pool partitioning and mobilization

Presynaptic vesicle pool availability is shaped by scaffold-dependent phosphosignalling pathways. Bassoon, an active zone scaffold described architecturally in Section 3.2.3, is essential for maintaining presynaptic phosphosignalling homeostasis. Its loss enhanced CDK5-dependent phosphorylation, hyperphosphorylating PDE4B at Ser145 to aberrantly increase cAMP hydrolysis and phosphorylating Synapsin I at Ser551 to strengthen F-actin binding, inducing reserve pool sequestration (Verstegen et al., 2014; Montenegro-Venegas et al., 2022). The consequent reduction in PKA activity shifted the PKA/calcineurin balance in favour of calcineurin, hypo-phosphorylating Synapsin I at site 1 and SNAP-25 at Thr138 (Section 3.3.1), silencing the synapse despite normal vesicle numbers. Pharmacological normalisation of CDK5, PDE4, or PKA activity with roscovitine, rolipram, or cAMP analogues respectively fully rescued these defects (Figure 5; Montenegro-Venegas et al., 2022).

Tomosyn1, a SNARE-binding protein, negatively regulates vesicle availability by controlling the partitioning between the recycling and resting pools. Its WD40 β -propeller domain bound Rab3A-GTP on synaptic vesicles and engaged Synapsin I, forming a Tomosyn1–Rab3A–Synapsin I retention complex (Cazares et al., 2016). CDK5-dependent phosphorylation of Tomosyn1 stabilised this complex at rest, while PKA-dependent phosphorylation at Ser724 reduced Tomosyn1 binding to syntaxin-1, facilitating SNARE complex formation and relieving its inhibitory brake on release (Baba et al., 2005). Upon increased activity, CaMKII- and PKA-mediated Synapsin I phosphorylation, together with Tomosyn1 Ser724 phosphorylation, weakened this retention and mobilised vesicles into the recycling pool. Consistent with this, Tomosyn1 knockdown increased recycling pool size and enhanced RRP release in cultured hippocampal neurons (Cazares et al., 2016).

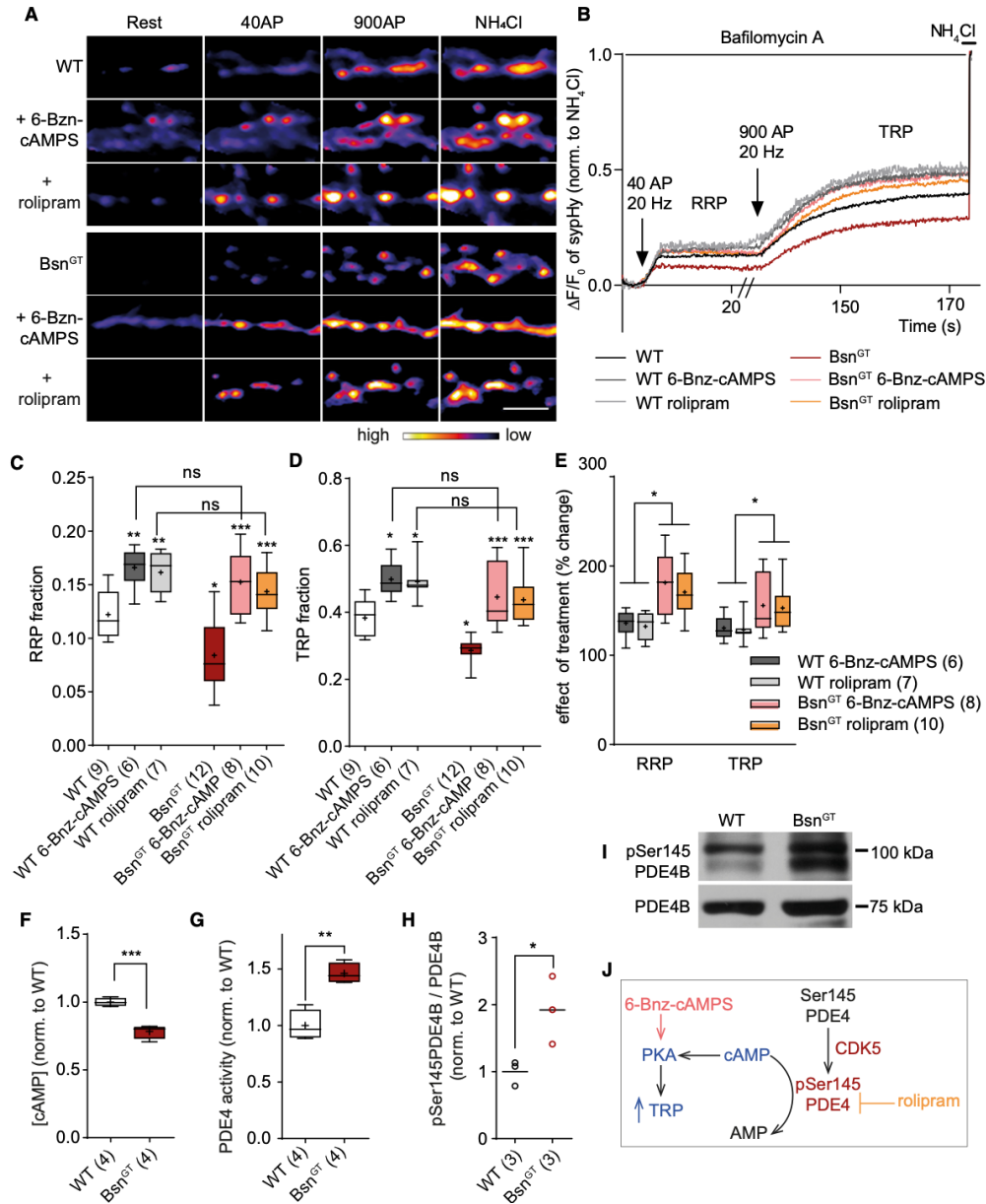


Figure 5. Bassoon maintains CDK5-PDE4 axis. (A-E) Quantitative imaging data demonstrating that the synaptic vesicle pool defects in Bassoon-deficient (Bsn^{GT}) neurons are PDE4-mediated. Treating neurons with either 6-Bnz-cAMPS (a cAMP analogue) or rolipram (a PDE4 inhibitor) fully restores the RRP and Total Recycling Pool (TRP) to wild type levels. (F-J) Biochemical measurements indicate reduced cAMP levels and elevated PDE4 enzymatic activity in Bsn^{GT} hippocampi. The Western blots in Panel I (quantified in H) reveal significantly higher phosphorylation of PDE4B at Ser145 in Bsn^{GT} tissue, a site phosphorylated by CDK5 to hyperactivate PDE4. (J) Summarises the cascade. (*p < 0.05; **p < 0.01; ***p < 0.001). Adapted from Montenegro-Venegas et al., 2022.

3.1.2. Phosphoregulation of presynaptic cytoskeleton and cytomatrix for vesicle mobilisation

Synapsin-dependent vesicle mobilisation is coupled to dynamic remodelling of the presynaptic actin cytoskeleton, which itself is regulated by phosphorylation. Actin-depolymerizing factor (ADF) and non-muscle cofilin (n-cofilin) are actin-regulatory proteins that sever F-actin, maintaining dynamic presynaptic actin turnover and synaptic vesicle mobilisation (Meng et al., 2002). Ca^{2+} -dependent cascade activated LIM kinase-1 (LIMK-1), which phosphorylated ADF/n-Cofilin on Ser3, inactivating its depolymerising activity. In mice lacking ADF/n-cofilin (ACC mice; $\text{ADF}^{-/-}/\text{n-Cof}^{\text{flx/flx, CamKII-cre}}$), the consequent accumulation of stable actin filaments impeded vesicle mobilisation from the reserve pool, resulting in a vesicle distribution shift away from the active zone. Functionally, the initial release probability and the number of docked vesicles were increased, followed by short-term depression as a consequence of impaired RRP replenishment and clearance (Wolf et al., 2015).

3.2. Docking and priming at the active zone: Phosphorylation-dependent control of release competence

Following mobilisation from the reserve pool, synaptic vesicles dock at the active zone and are primed into a fusion-competent state by the coordinated actions of Munc18-1, Munc13-1, syntaxin-1 and RIM1. Munc18-1 bound syntaxin-1 in its closed conformation, stabilising it at the plasma membrane and preventing premature SNARE assembly (Dulubova et al., 1999; Yang et al., 2000; Arunachalam et al., 2008). Munc13-1 was initially autoinhibited by C2A homodimerization. RIM disrupted this dimer to form an active RIM–Munc13-1 heterodimer (Dulubova et al., 2005; Lu et al., 2006; Camacho et al., 2017). Activated Munc13-1 then promoted syntaxin-1 opening through its MUN domain, enabling SNARE complex assembly with SNAP-25 and synaptobrevin (Ma et al., 2011), with Munc18-1 and complexin-1 remaining associated to stabilise the primed state (Deák et al., 2009). Phosphoproteomic analyses confirmed that these SNARE-associated proteins undergo activity-dependent phosphorylation, providing a mechanism to dynamically regulate presynaptic plasticity (Kohansal-Nodehi et al., 2016).

3.2.1. Phosphorylation of the Munc18-1–syntaxin-1 priming module in presynaptic plasticity

Munc18-1 is an essential phosphoprotein of neurotransmitter release whose deletion causes widespread neurodegeneration (Verhage et al., 2000). PKC phosphorylation at Ser306 and Ser313 reduced syntaxin-1 affinity (Fujita et al., 1996; Barclay et al., 2003), with context-dependent functional consequences. In hippocampal autaptic neurons, PKC-insensitive Munc18-1 (S306A/S313A; alanine substitutions denote phosphodead mutants and aspartate/glutamate substitutions denote phosphomimetic mutants; Appendix 4) supported basal transmission, RRP size, and release probability, but abolished DAG/phorbol ester potentiation and disrupted activity-dependent vesicle redistribution toward release sites (Wierda et al., 2007). At the calyx of Held, PKC phosphorylation of Munc18-1 contributed to PTP, and inhibiting dephosphorylation with calyculin prolonged PTP duration (Figure 6; Genç et al., 2014). PTP nonetheless persisted at CA3-CA1 and parallel fibre-Purkinje cell synapses in Munc18-1SA knock-in mice lacking PKC sites (S306/S312/S313) (Wang et al., 2021).

On the other hand, ERK phosphorylated Munc18-1 at Ser241 within the closed syntaxin-1 binding pocket to induce longer-lasting presynaptic inhibition. CB1 receptors, group II/III mGluRs, and TrkB signalling each increased Ser241 phosphorylation via ERK activation in hippocampal neurons (Schmitz et al., 2016). Phosphomimetic S241D depressed synaptic strength and triggered ubiquitin-proteasome-dependent degradation of Munc18-1, whereas phospho-blocking S241A enhanced EPSC amplitude, RRP size, and mEPSC frequency. Accordingly, blocking Ser241 phosphorylation or ERK activity attenuated CB1- and mGluR2/3-dependent presynaptic inhibition and depolarisation-induced suppression of excitation, identifying Ser241 as an effector site for neuromodulator-triggered depression (Schmitz et al., 2016).

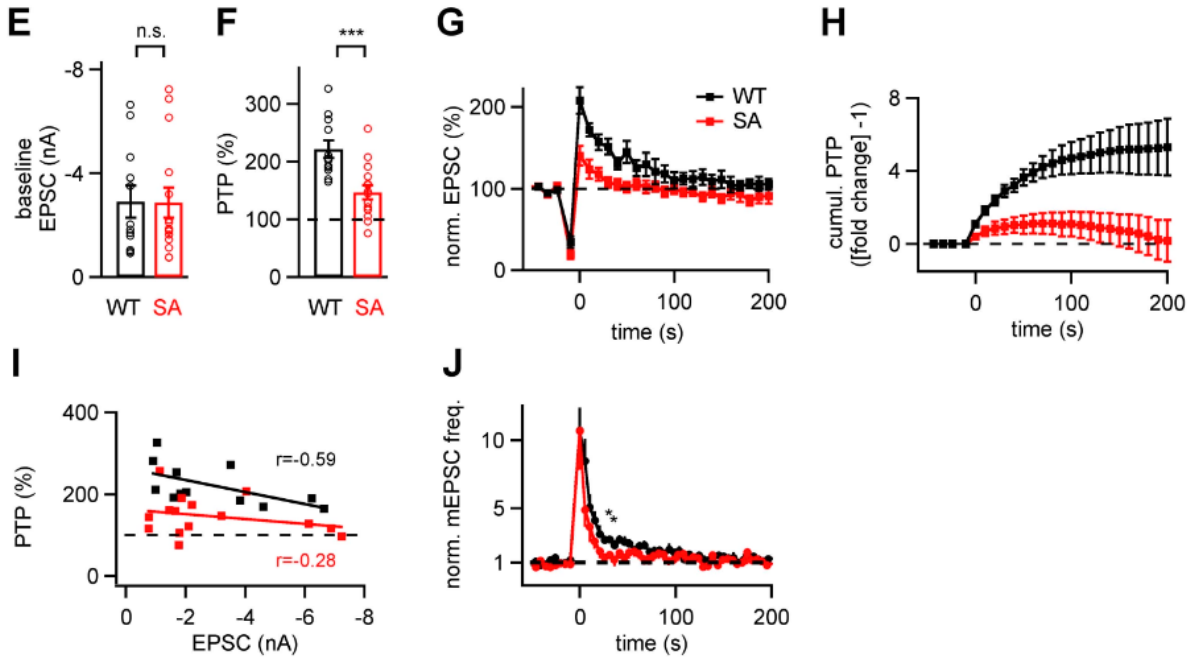


Figure 6. PKC phosphorylation of Munc18-1 is required for post-tetanic potentiation (PTP). (E-H) Summary data from calyx of Held synapses rescued with wild-type Munc18-1 (WT) or the PKC-insensitive mutant M18^{SA} (S306A, S312A, S313A). (E) Baseline EPSC amplitude is unchanged between WT and M18^{SA} synapses, indicating that PKC phosphorylation is not required for basal evoked release. (F) Peak PTP is significantly reduced in M18^{SA} synapses. (G) Time course of normalized EPSC amplitudes after tetanic stimulation shows robust potentiation in WT synapses but attenuated potentiation in M18^{SA} synapses. (H) Cumulative PTP is likewise markedly reduced in M18^{SA} synapses, indicating that PKC phosphorylation of Munc18-1 contributes to the magnitude and persistence of post-tetanic enhancement of neurotransmitter release. (**p < 0.01, ***p < 0.001). Adapted from Genç et al., 2014.

3.2.2. RIM1 as a phosphoregulated organiser of active-zone architecture and presynaptic plasticity

Rab3-interacting molecule 1 (RIM1) is a core active-zone scaffold that organises release sites by coordinating vesicle recruitment, docking, and coupling to Ca^{2+} entry. It clustered Ca^{2+} channels at active zones (Han et al., 2011; Kaeser et al., 2011) and, through a tripartite Rab3–RIM–Munc13 complex, tethered vesicles adjacent to the priming machinery (Dulubova et al., 2005). Consistent with this central role, deletion of RIM1 α abolished presynaptic LTP at hippocampal mossy fibre and cerebellar parallel fibre synapses (Castillo et al., 2002).

RIM1 is extensively phosphoregulated. Lonart et al. (2003) showed that, in cultured cerebellar granule cell-Purkinje neuron synapses derived from RIM1 α knockout mice, presynaptic expression of wild-type RIM1 α rescued LTP, whereas a Ser413A mutant failed to do so.

However, follow up in vivo knock-in studies revealed that mutation of Ser413 did not abolish presynaptic LTP at mossy fibre or parallel fibre synapses, nor did it impair learning (Kaeser et al., 2008). Recently, Müller et al. (2022) identified another phosphosite, Ser1045 (S1045), to mediate RIM1 nanocluster formation for presynaptic homeostatic plasticity. Chronic synaptic silencing activated serine/arginine protein-specific kinase 2 (SRPK2), which phosphorylated RIM1 at S1045, promoting the formation of RIM1 nanoclusters within the active zone, increasing the local density of release sites and the number of docked synaptic vesicles. Phosphomimetic RIM1 mutant (S1045E) enhanced neurotransmitter release (Figure 7).

Furthermore, RIM1-dependent release can also be strengthened indirectly through phosphorylation of its binding partners. CDK5 phosphorylated the serine residues of CaV2.2 within its RIM-binding domain, enhancing the CaV2.2–RIM1 interaction, channel open probability, and Ca^{2+} influx. These effects were lost in a phospho-dead mutant (Figure 8). Conversely, CDK5 inhibition or disruption of the phospho-specific epitope reduced neurotransmitter release and docked vesicle number (Su et al., 2012).

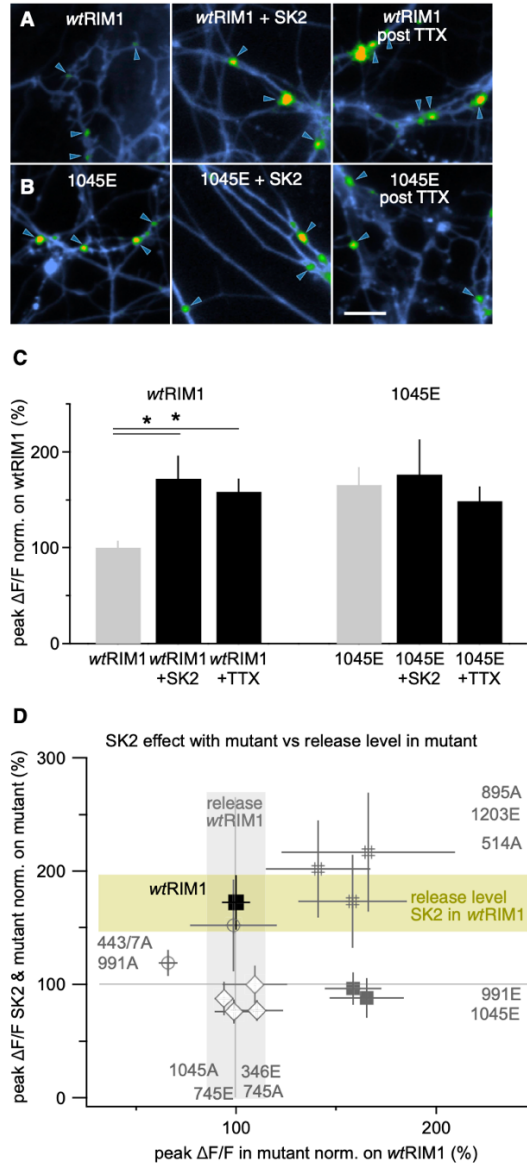


Figure 7. Phosphorylation of RIM1 at Ser1045 enhances presynaptic release during homeostatic plasticity. (A-B) Representative images of presynaptic release sites in neurons expressing wild-type RIM1 or the phosphomimetic mutant RIM1^{S1045E}. Overexpression of SRPK2 (SK2) or synaptic silencing with TTX increases presynaptic signal intensity in wtRIM1 neurons, whereas S1045E displays elevated release that is not further enhanced by these manipulations. Arrowheads indicate analysed presynaptic sites. (C) Quantification of peak $\Delta F/F$ shows that SRPK2 expression or TTX treatment significantly increases release in wtRIM1 neurons but has little effect in S1045E neurons, indicating occlusion of potentiation. (D) Mutational analysis of RIM1 phosphosites identifies Ser1045 as a critical site required for SRPK2-dependent enhancement of release. Adapted from Müller et al., 2022.

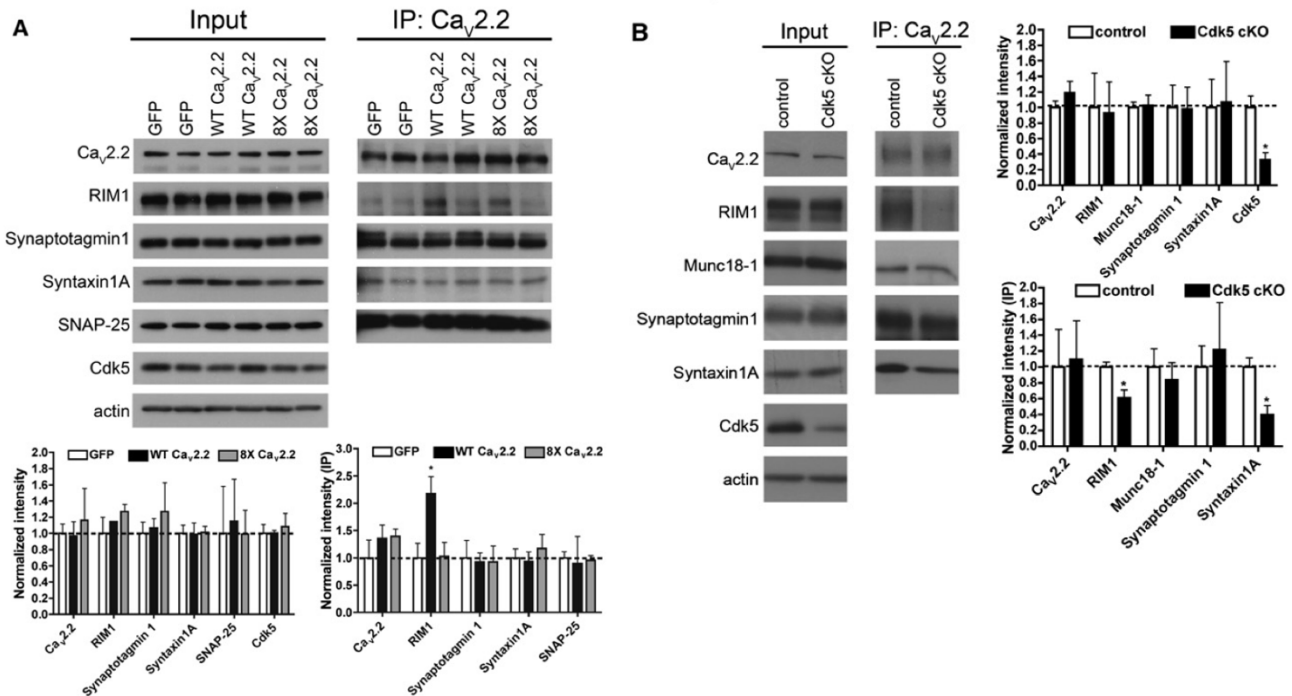


Figure 8. CDK5 phosphorylation of CaV2.2 enhances its interaction with RIM1.

(A) Co-immunoprecipitation of CaV2.2 complexes from cells expressing GFP, wild-type CaV2.2, or a phosphorylation-deficient 8× CaV2.2 (eight serine residues mutated to alanine) shows increased association of RIM1 with wild-type channels, which is lost in the mutant. Other presynaptic proteins are unchanged. (B) Co-immunoprecipitation from control and CDK5 conditional KO brains shows reduced association of CaV2.2 with RIM1 and Munc18-1 after CDK5 loss, with little change in other interactions. These data indicate that CDK5 phosphorylation promotes coupling of CaV2.2 to RIM1-containing active-zone complexes. (**p* < 0.05). Adapted from Su et al., 2012.

3.2.3. Bassoon as a phosphoregulated stabiliser of active zone architecture and presynaptic plasticity

Bassoon is a large active zone cytomatrix scaffold that undergoes extensive phosphoregulation, with at least 48 phosphosites dynamically modulated during Ca²⁺-dependent exocytosis (Kohansal-Nodehi et al., 2016). It bound RIM-binding protein (RBP) via a C-terminal PXXP motif to recruit CaV2.1 channels to release sites. The loss of Bassoon reduced synaptic RBP2 and CaV2.1 levels, impaired CaV2.1-dependent exocytosis, and shifted release towards CaV2.2

(Davydova et al., 2014). This architectural role is dynamically tuned by phosphoregulation: Ribosomal S6 kinase (RSK)-mediated phosphorylation at Ser2845 promoted 14-3-3 binding, facilitating Bassoon anchoring and synaptic turnover. The molecular exchange was decreased in the phospho-dead S2845A mutant (reduced FRAP recovery; Figure 9; Appendix 5; Schröder et al., 2013).

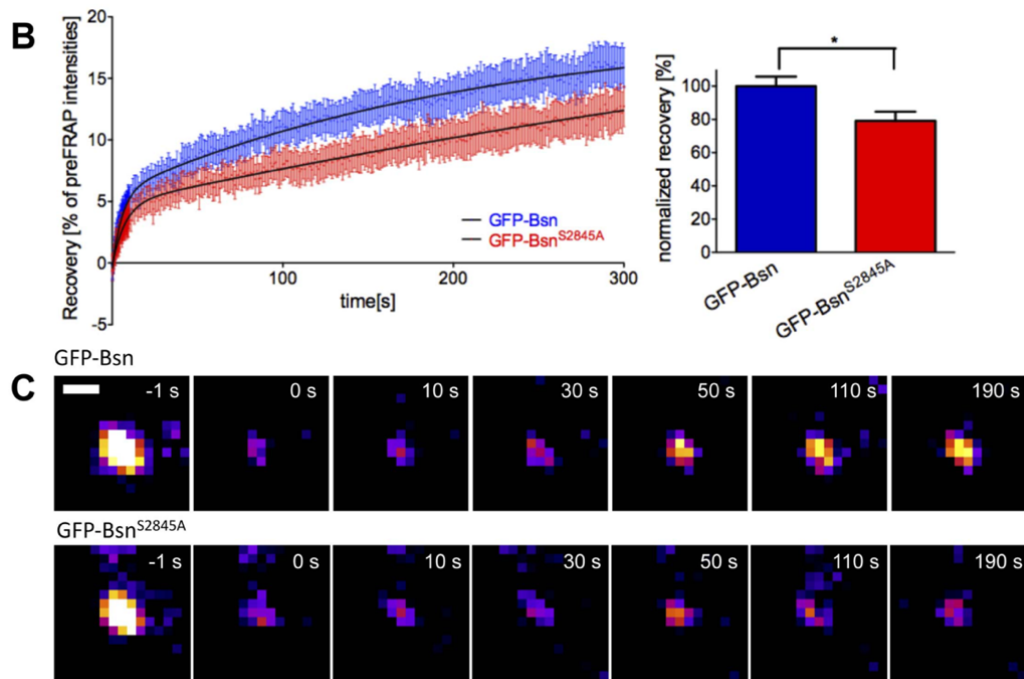


Figure 9. Phosphorylation-dependent regulation of Bassoon mobility at presynaptic sites. (B) FRAP recovery curves and quantification for GFP-Bassoon (GFP-Bsn) and GFP-Bsn^{S2845A} show significantly reduced recovery of the S2845A mutant, indicating decreased molecular exchange at synapses. (C) Representative FRAP images illustrate slower fluorescence recovery of GFP-Bsn^{S2845A} compared with wild-type Bassoon following photobleaching. These findings support a role for Ser2845 phosphorylation in promoting Bassoon mobility at the active zone. Adapted from Schröder et al., 2013.

3.3. Fusion competence: Phosphorylation-dependent regulation of Ca^{2+} -triggered release

The neuronal SNARE complex forms the core machinery for Ca^{2+} -triggered synaptic vesicle fusion. It is a stable four- α -helix bundle comprising one helix from synaptobrevin, one from syntaxin-1, and two from SNAP-25 (Sutton et al., 1998). The complex assembled by N- to C-terminal zippering: N-terminal assembly contributed to vesicle docking and priming, whereas progressive C-terminal zippering generated the mechanical force required to bring the vesicular and plasma membranes together (Li et al., 2007; Gao et al., 2012). Fusion competence is then further tuned by phosphorylation-dependent regulation of the Ca^{2+} sensor, fusion clamp, and associated SNARE machinery.

3.3.1. SNARE Complex Modulation by SNAP-25 and Snapin Phosphorylation

SNAP-25 was phosphorylated at Ser187 (PKC) and Thr138 (PKA). In chromaffin cells, PKC-dependent phosphorylation at Ser187 (S187E) enhanced SNAP-25 interactions with syntaxin-1 and increased vesicle priming and the size of highly Ca^{2+} -sensitive pool (HCSP) (Nagy et al., 2002; Sørensen et al., 2003; Yang et al., 2007). At the neuromuscular junction, presynaptic M_2 mAChR signalling reduced PKC ϵ expression, decreasing Ser187 phosphorylation for homeostatic control of acetylcholine release (Polishchuk et al., 2025). However, the contribution of Ser187 phosphorylation to phorbol-ester/ Ca^{2+} -dependent potentiation is synapse-dependent, being dispensable in hippocampal CA3 synapses and glutamatergic autapses (Finley, Scheller and Madison, 2003; Ruiter et al., 2020).

PKA phosphorylation of SNAP-25 at Thr138 regulated releasable pool size, as the phospho-dead mutation (T138A) reduced the primed vesicle pool (Hepp et al., 2002; Nagy et al., 2004). PKA also targeted the accessory protein Snapin. In hippocampal slices, phosphorylation at Ser50 increased Snapin affinity for SNAP-25 and enhanced synaptotagmin–SNARE complex association (Chheda et al., 2001). Consistently, phosphomimetic Snapin (S50D) in cultured hippocampal neurons increased vesicle release probability and accelerated synaptic depression during high-frequency stimulation, reducing RRP (Thakur et al., 2004). Snapin S50D overexpression in adrenal chromaffin cells, on the other hand, increased the number of release-competent vesicles (Chheda et al., 2001).

3.3.2. Phosphorylation of Synaptotagmin isoforms regulates release

Synaptotagmin-1 (Syt1) is the principal vesicular Ca^{2+} sensor for fast synchronous neurotransmitter release, coupling Ca^{2+} entry via its two C2 domains to the SNARE machinery for synchronised exocytosis. Syt1 was phosphorylated at Thr112 within its juxtamembrane linker by PKC and CaMKII (Hilfiker et al., 1999). In hippocampal neurons, introducing T112A mutation abolished DAG/PKC-dependent potentiation of evoked release without affecting basal transmission, PPF, vesicle priming, or RRP size (Figure 10; de Jong et al., 2016). Deletion of the 109-116 linker segment, or substitution with synaptotagmin-2, which lacks this sequence, supported normal or enhanced potentiation without requiring phosphorylation, likely indicating its inhibitory role (de Jong et al., 2016). Phosphorylation-dependent control also extends to longer-lasting forms of presynaptic plasticity via synaptotagmin-12 (Syt12), a Ca^{2+} -independent synaptic vesicle isoform that is phosphorylated by PKA at Ser97. Kaeser-Woo et al. (2013) demonstrated that Syt12 phosphorylation by PKA was essential for cAMP-dependent hippocampal mossy fibre LTP, as both Syt12 KO and Syt12 S97A KI mice exhibited impaired mossy fibre LTP.

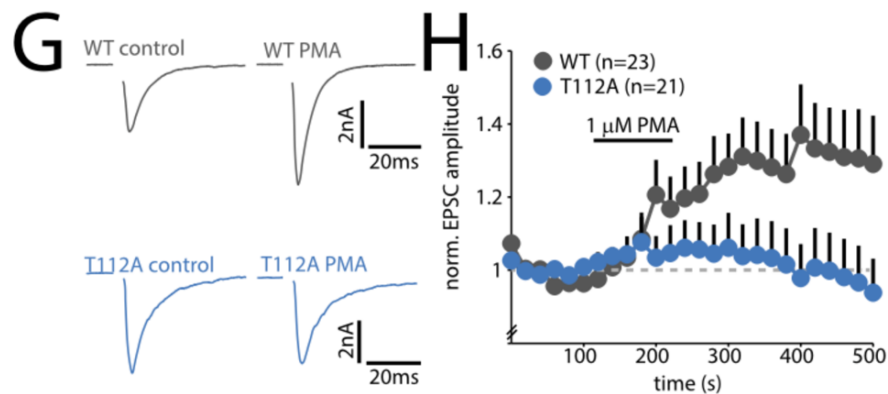


Figure 10. Syt1 Thr112 is required for DAG/PKC-dependent potentiation of evoked neurotransmitter release. (G) Representative evoked EPSC traces from hippocampal neurons expressing Syt1WT or the non-phosphorylatable mutant Syt1T112A before and after phorbol ester (PMA) application. (H) PMA-induced potentiation of evoked release is markedly reduced in Syt1T112A, as shown by quantification of EPSC enhancement, indicating that the Thr112 site is required for DAG/PKC-pathway-dependent presynaptic potentiation. Adapted from de Jong et al., 2016.

3.4. Post-fusion events: phosphorylation-dependent control of endocytosis and release-site reuse

The nerve terminal contains a limited number of recycling vesicles relative to the high rate of exocytosis required for neurotransmission, making rapid vesicle retrieval necessary for sustained synaptic transmission and implicating recycling kinetics in short-term depression and recovery of synaptic strength (Alabi and Tsien, 2012; Watanabe et al., 2013). A calcineurin-CDK5 phosphocycle coordinates this post-fusion recovery (Cousin and Robinson, 2001; Tan et al., 2003).

3.4.1. Calcineurin-Cdk5 phosphocycling controls assembly of the endocytic machinery

During synaptic vesicle endocytosis (SVE) (Appendix 6), dynamin I, a large GTPase required for membrane fission, assembled at the neck of invaginated membranes where GTP hydrolysis induced vesicle scission (Ferguson et al., 2007). In resting terminals, dynamin I and other endocytic dephosphins were predominantly phosphorylated. Upon stimulation, Ca^{2+} influx activated calcineurin, which dephosphorylated dynamin I at Ser774 and Ser778, promoting the formation of a dynamin I-syndapin I complex that facilitated vesicle retrieval, particularly during intense activity-dependent bulk endocytosis (Tan et al., 2003). Mutation of these sites to alanine arrested SVE in cultured neurons (Anggono et al., 2006). Following retrieval, CDK5 rephosphorylated endocytic proteins, resetting the machinery for subsequent rounds of recycling (Tan et al., 2003). Downstream of vesicle fission, calcineurin-dependent dephosphorylation of synaptojanin 1 enhanced its phosphatase activity, promoting PI(4,5)P₂ hydrolysis and thereby weakening clathrin adaptor-membrane interactions to facilitate vesicle uncoating (Cousin and Robinson, 2001; Lee *et al.*, 2004). Consistent with this, synaptojanin 1-KO mice showed nerve terminal accumulation of clathrin-coated vesicles and delayed recovery, resulting in neurological defects and postnatal lethality (Cremona *et al.*, 1999).

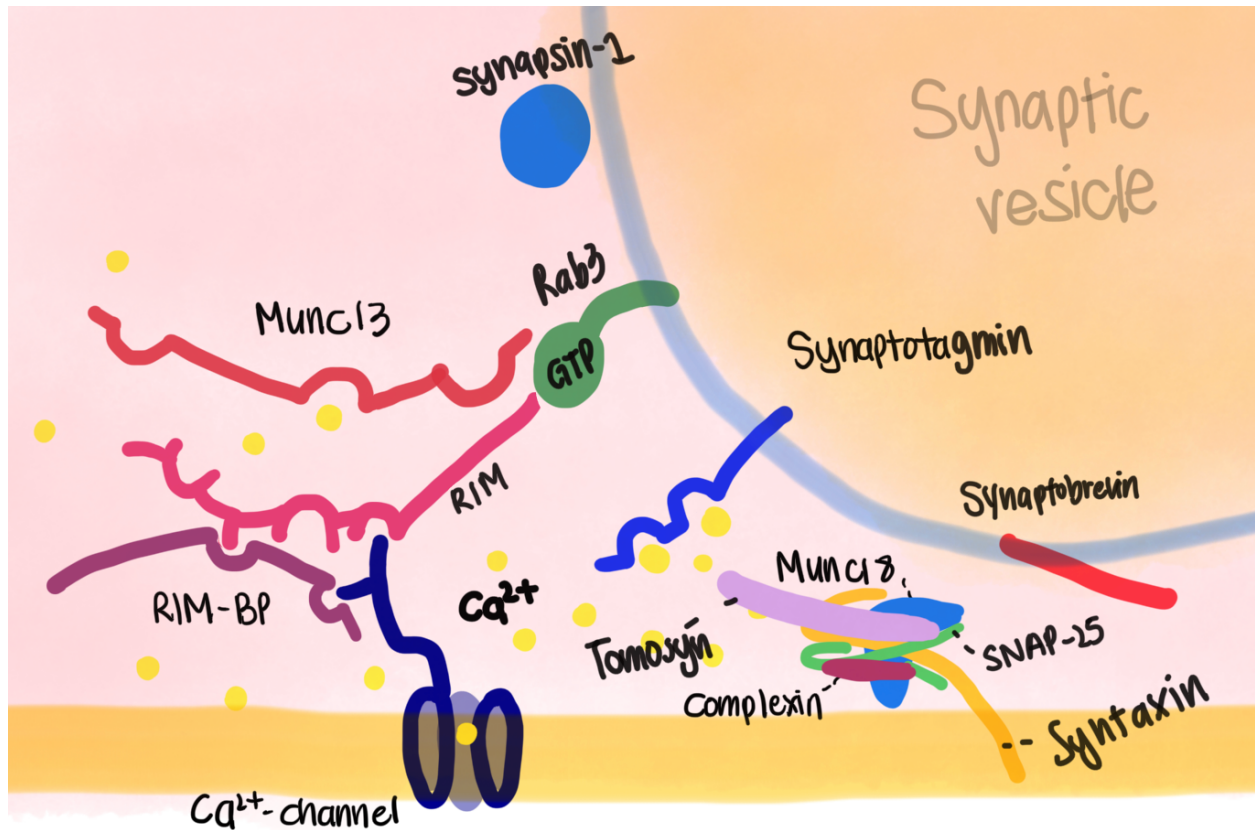


Figure 11. Integrated organisation of the phosphoregulated presynaptic release machinery across the synaptic vesicle cycle. Schematic of a synaptic vesicle and active zone showing major regulators of vesicle availability, docking, priming and Ca^{2+} -triggered fusion, including Synapsin I, Rab3, RIM, Munc13, RIM-BP, Tomosyn, synaptotagmin, complexin, Munc18-1 and the core SNARE proteins. Modified from Kaeser et al., 2011.

Protein	Phosphosite	Kinase / Phosphatase	Functional Effect	Timescale	Plasticity Type
3.1 Vesicle Mobilisation & Cytoskeletal Regulation					
Synapsin I	Site 1 (Ser9)	PKA	Disperses condensates; releases vesicles from reserve pool	ms-s	STP (augmentation, PTP)
	Sites 2 & 3 (Ser566/603)	CaMKII	Disperses condensates; promotes reserve to recycling pool transition	ms-s	STP (augmentation, PTP)
	Ser551	CDK5	Strengthens F-actin binding; enforces reserve pool sequestration	Tonic	Constrains basal STP
	Sites 4-6 (MAPK sites)	Calcineurin (dephosphorylation)	Dephosphorylation limits glutamate release	s-min	STP (depression)
Tomosyn1	Ser724	PKA / CaMKII	Disrupts Tomosyn1-Rab3A-Synapsin complex; facilitates SNARE assembly	ms-s	STP (facilitation)
PDE4B	Ser145	CDK5	Hyperactivates PDE4; collapses cAMP/PKA → silences vesicle pools	Tonic	Homeostatic
ADF/n-cofilin	Ser3	LIMK-1	Inactivates F-actin severing; impairs vesicle mobilisation to AZ	ms-s	STP (depression↑, RRP replenishment↓)
3.2 Docking, Priming & Active Zone Architecture					
Munc18-1	Ser306 / Ser313	PKC	Reduces syntaxin-1 affinity; enhances release and contributes to PTP	min	STP (PTP)
	Ser241	ERK	Suppresses synaptic strength; triggers proteasomal degradation	min-h	Neuromodulatory depression
SNAP-25	Thr138	PKA	Maintains primed vesicle pool size	s-min	STP (pool regulation)
	Ser187	PKC	Enhances syntaxin-1 interaction; increases HCSP size (chromaffin)	s-min	STP (synapse-type specific)
Snapiin	Ser50	PKA	Increases SNAP-25 affinity; raises vesicle P_r ; accelerates depression	s-min	STP (depression)
RIM1 α	Ser413	PKA	Disputed: rescue studies suggest LTP role; knock-in shows dispensable	h	Presynaptic LTP (contested)

	Ser1045	SRPK2	Promotes RIM1 nanocluster formation; scales release site number	h	Homeostatic plasticity
CaV2.2	Multiple Ser (RIM-binding domain)	CDK5	Enhances CaV2.2-RIM1 coupling; increases Ca ²⁺ influx and docking	Tonic	Basal Pr, STP
Bassoon	Ser2845	RSK	Promotes 14-3-3 binding; regulates scaffold mobility/turnover at AZ	Tonic	Active zone maintenance
3.3 Ca²⁺-Triggered Fusion Competence					
Synaptotagmin-1 (Syt1)	Thr112	PKC / CaMKII	Required for DAG/PKC-dependent potentiation; spares basal transmission	min	STP (PKC-dependent PTP)
Synaptotagmin-12 (Syt12)	Ser97	PKA	Essential for cAMP-dependent mossy fibre LTP	h	Presynaptic LTP
3.4 Endocytosis & Release-Site Recovery					
Dynamin I	Ser774 / Ser778	Calcineurin (dephosphorylation) CDK5 (rephosphorylation)	Dephosphorylation enables dynamin-syndapin complex; drives SVE	ms-s	STP (recovery from depression)
Synaptojanin	Ser1144	Calcineurin (dephosphorylation)	PI(4,5)P ₂ hydrolysis facilitates clathrin uncoating post-fission	ms-s	STP (vesicle recycling)

Table 1. Phosphoregulation of the synaptic vesicle cycle: substrates, kinases, and plasticity outcomes. Major phosphoproteins are organised by functional stage – vesicle mobilisation (Section 3.1), docking and priming (Section 3.2), fusion competence (Section 3.3), and endocytosis (Section 3.4) – with their relevant phosphosite(s), responsible kinase or phosphatase, functional consequence, timescale, and associated plasticity type indicated.

4. Discussion

4.1. Synthesis: Phosphorylation as a unifying mechanism of presynaptic plasticity

Phosphorylation constitutes a unifying regulatory mechanism underlying presynaptic plasticity, offering a multi-site, bidirectional code for precise, activity-dependent control of synaptic output across multiple timescales (Table 1; Figure 11). A defining feature of phosphoregulation is the convergence of opposing kinase signals onto shared substrates. PKA and CaMKII promote Synapsin I-dependent vesicle mobilisation, while CDK5 promotes reserve pool sequestration via the same protein (Chi, Greengard and Ryan, 2003; Versteegen et al., 2014). Similarly, PKC phosphorylation of Munc18-1 at Ser306/313 enhances release, while ERK-dependent phosphorylation at Ser241 suppresses synaptic strength and accelerates its degradation (Wierda et al., 2007; Schmitz et al., 2016). This bidirectionality demonstrates that kinase balance, rather than the activity of any single enzyme, encodes presynaptic output.

Addressing the three objectives of this dissertation, distinct phosphorylation pathways selectively modulate plasticity across timescales. On millisecond-to-second timescales, rapid phosphorylation of Synapsin I (Ser566/603; Milovanovic et al., 2018), ADF/n-cofilin, and the endocytic dephosphins (dynamin I Ser774/778; Tan et al., 2003) modulates vesicle availability, cytoskeletal dynamics, and endocytic recovery – adjusting both the effective vesicle pool and replenishment rate during STP (Cousin and Robinson, 2001). On the timescale of minutes, PKC-dependent phosphorylation of Munc18-1 Ser306/313 and Syt1 Thr112 tunes the priming clamp and Ca^{2+} sensor to underlie potentiation following tetanic stimulation (Genç et al., 2014; de Jong et al., 2016). On the timescale of hours, PKA-mediated phosphorylation of RIM1 α and Syt12 Ser97 sustains enduring increases in P_r and N – through nanocluster remodelling (RIM1 α Ser1045; Müller et al., 2022) and Ca^{2+} channel coupling (CaV2.2; Su et al., 2012; Bassoon Ser2845; Davydova et al., 2014) – defining presynaptic LTP at mossy fibre-CA3 synapses (Castillo et al., 2002; Kaeser-Woo et al., 2013). Homeostatic stabilisation is mediated through SRPK2-RIM1 nanocluster scaling and the Bassoon-CDK5-PDE4B-PKA cascade, which together demonstrate that phosphosignalling state, not vesicle number, determines how many

anatomically docked vesicles are release-competent – rendering morphological counts alone insufficient as a proxy for functional pool size (Montenegro-Venegas et al., 2022; Müller et al., 2022).

4.2. Limitations

Three methodological limitations recur across this literature. First, many conclusions rely on phosphomimetic or phosphodead substitutions that approximate but do not replicate authentic phosphorylation (Pérez-Mejías et al., 2020; Kozeleková et al., 2022; Appendix 4) and should therefore be considered indicative rather than definitive pending corroboration from acute, kinase-specific manipulations. Second, the RIM1 α Ser413 discrepancy illustrates that phosphosite necessity assigned in reduced preparations can be overturned by endogenous knock-in analysis in vivo (Lonart et al., 2003; Kaeser et al., 2008), underscoring that causal responsibility requires in vivo validation. Third, conclusions from non-specific or heterologous systems require caution – the global LIMK-1 knockout cannot resolve presynaptic from postsynaptic contributions to STP (Meng et al., 2002; Wolf et al., 2015), and divergent Snapin Ser50 effects across hippocampal neurons and chromaffin cells likely reflect differences in vesicle pool organisation – large dense-core vesicles distributed across multiple pools in chromaffin cells versus a single RRP in hippocampal synapses – rather than a true mechanistic discrepancy (Chheda et al., 2001; Thakur et al., 2004).

4.3. Future Direction

Although Kohansal-Nodehi et al. (2016) demonstrated widespread simultaneous phosphorylation and dephosphorylation across presynaptic proteins during exocytosis, their bulk synaptosome approach cannot resolve whether specific phosphosite combinations are necessary or sufficient for a given plasticity outcome at individual boutons. Future work should combine single-synapse phosphoproteomics with spectrally separable genetically encoded kinase activity reporters for PKA, CaMKII, and CDK5, paired with site-specific phosphonull knock-in lines, to establish whether co-occurring phosphorylation events operate cooperatively or redundantly to gate vesicle pool dynamics.

Furthermore, the mechanistic basis of presynaptic LTP at mossy fibre-CA3 synapses remains unresolved. The RIM1 α Ser413 discrepancy – dispensable in vivo (Kaesler et al., 2008) despite being implicated by in vitro rescue (Lonart et al., 2003) – may be a consequent compensation by Syt12 Ser97 phosphorylation (Kaesler-Woo et al., 2013). A compound knock-in simultaneously ablating both sites, assessed by direct voltage-clamp at identified mossy fibre boutons, would establish which phosphorylation event is necessary and sufficient for presynaptic LTP induction.

4.4. Conclusion

Phosphorylation-dependent modulation of the presynaptic terminal is multifaceted: the convergence of opposing kinase pathways onto shared substrates, rather than the activity of any single kinase, encodes the direction and magnitude of plasticity across timescales. Critically, the finding that phosphosignalling state – not vesicle number – determines release competence reframes how synaptic dysfunction in disease should be interpreted, since morphological preservation is not functional preservation. Understanding this phosphorylation-dependent logic in sufficient molecular detail to predict its dysregulation, and to design phosphosite-targeted therapeutic strategies that restore normal presynaptic gain without disrupting the broader signalling architecture of the synapse, represents one of the most compelling frontiers in cellular neuroscience and pharmacology.

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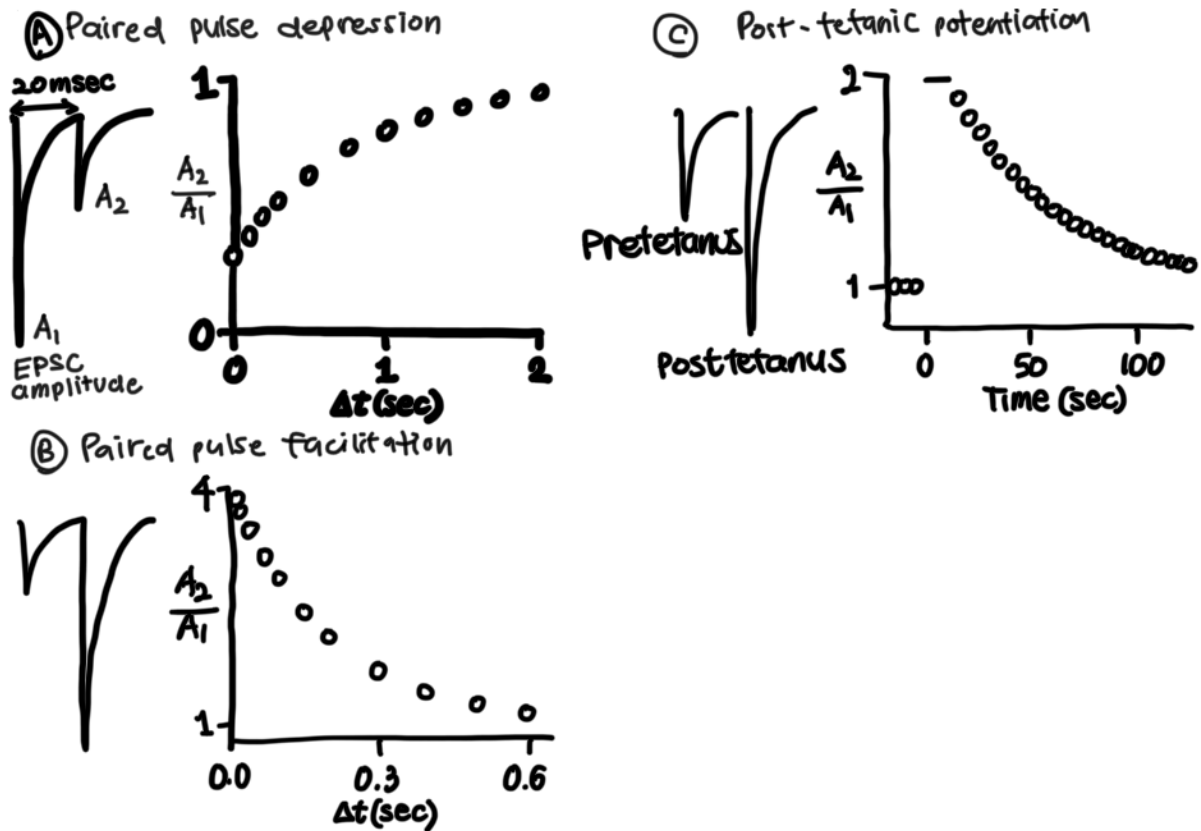
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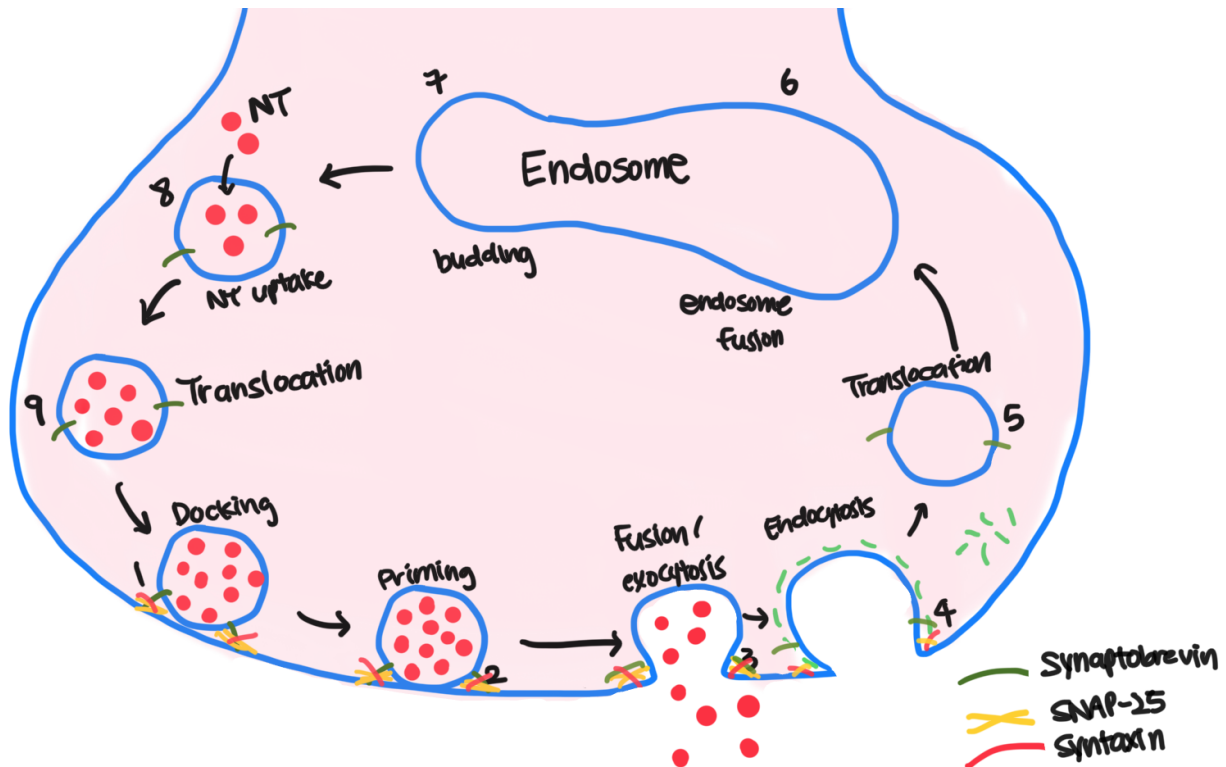
6. Appendix

6.1. Appendix 1: Major forms of short-term plasticity



Major forms of short-term synaptic plasticity. Diagram showing four major forms of use-dependent presynaptic plasticity. (A) Paired-pulse depression is typically caused by rapid depletion of fusion-competent vesicles or transient inactivation of release sites following the first stimulus (B) Paired-pulse facilitation is mainly driven by residual Ca^{2+} that persists between closely spaced stimuli and transiently increases release probability. (C) Augmentation and post-tetanic potentiation result from prolonged presynaptic Ca^{2+} accumulation and activation of signalling mechanisms that enhance release probability, vesicle priming, or the effective readily releasable pool (RRP) (Data from Regehr, 2012).

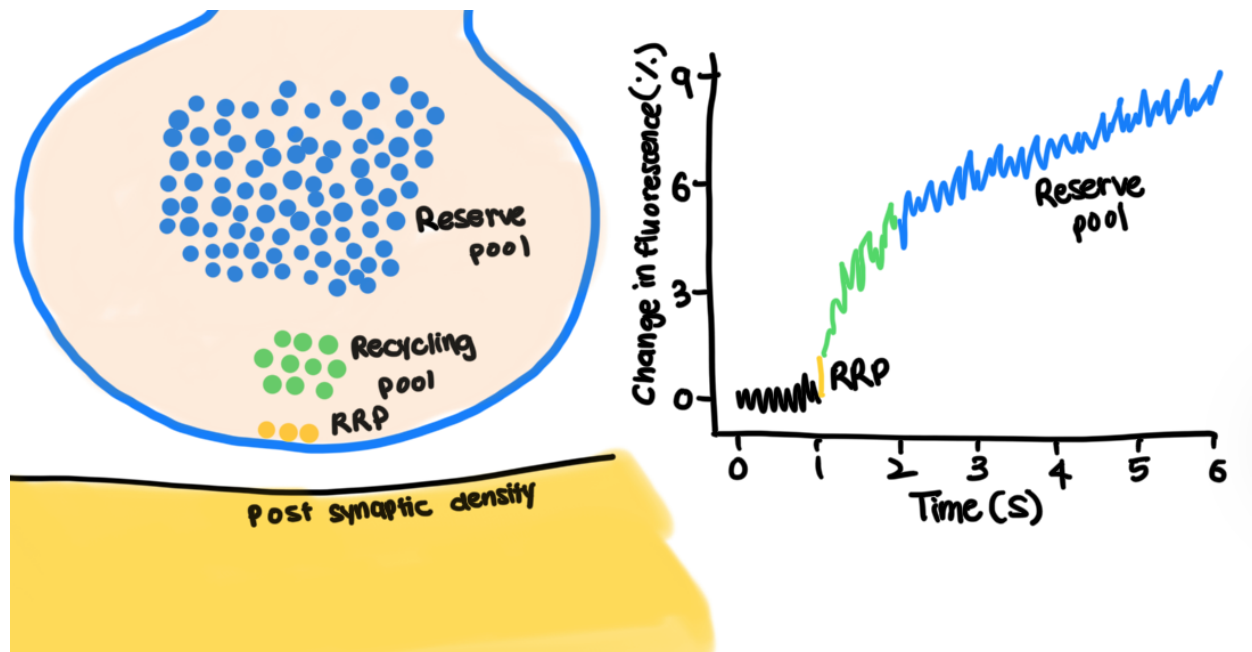
6.2. Appendix 2: The synaptic vesicle cycle



The synaptic vesicle cycle in the presynaptic terminal.

Schematic overview of the nine sequential steps of the synaptic vesicle (SV) cycle at the nerve terminal: (1) docking of transmitter-filled vesicles at the active zone, (2) priming to render docked vesicles fusion-competent, (3) Ca^{2+} -triggered fusion/exocytosis, (4) endocytosis of vesicle membrane after release, (5) translocation of retrieved vesicles into the terminal interior, (6) fusion with endosomal intermediates, (7) budding/regeneration of new synaptic vesicles, (8) neurotransmitter uptake into reformed vesicles, and (9) translocation of refilled vesicles back to the active zone for another round of release. This figure summarizes the concept that synaptic transmission depends on coordinated coupling between rapid exocytosis and local vesicle recycling. Modified from Südhof, 2004.

6.3. Appendix 3: Organisation of synaptic vesicle pools



Functional organisation of synaptic vesicle pools at the presynaptic terminal.

Synaptic vesicles are classically divided into three functional pools: the readily releasable pool (RRP), comprising vesicles docked and primed at or very near the active zone and therefore immediately available for fusion; the recycling pool, which sustains neurotransmitter release during ongoing physiological stimulation; and the reserve pool, which is recruited mainly during intense or prolonged activity. Although the RRP is closely associated with release sites, vesicle pools are defined primarily by their release competence and mobilisation kinetics, rather than by strict anatomical segregation, and recycling and reserve pool vesicles may be spatially intermixed within the bouton. Modified from Rizzoli and Betz (2005).

6.4. Appendix 4: Modelling phosphorylation in proteins

In laboratory studies, phosphorylation is commonly modelled by site-directed mutagenesis of the amino acid residue that normally accepts the phosphate group. For serine or threonine phosphorylation sites, substitution with alanine is typically used to generate a non-phosphorylatable or phosphodead/phosphonull mutant, because alanine lacks the hydroxyl group required for phosphate attachment. By contrast, substitution with aspartate or glutamate is often used to create a phosphomimetic mutant, as these acidic residues introduce a constitutive

negative charge that can approximate some electrostatic consequences of phosphorylation. This strategy allows investigators to compare the behaviour of a protein in putative non-phosphorylated and phosphorylation-like states, thereby inferring how modification at a given residue influences protein interactions, localisation, stability or function. However, phosphomimetic substitutions are only an approximation: a true phosphate group differs from Asp/Glu in size, charge distribution and hydrogen-bonding potential, meaning that phosphomimetic mutants do not always reproduce the structural or functional effects of authentic phosphorylation. Accordingly, results from phosphodead and phosphomimetic mutants should be interpreted as useful experimental proxies rather than exact replicas of the native phosphorylated state (Pérez-Mejías et al., 2020; Kozeleková et al., 2022).

6.5. Appendix 5: Fluorescence recovery after photobleaching (FRAP)

FRAP is used to assess the mobility of fluorescently tagged proteins within a defined subcellular region. After local photobleaching of synaptic Bassoon fluorescence, recovery reflects the exchange of unbleached Bassoon molecules into the bleached area. Rapid or extensive recovery indicates a more mobile, dynamically exchanging scaffold population, whereas reduced recovery indicates tighter anchoring and lower molecular turnover. Thus, in the context of Bassoon, decreased FRAP recovery after preventing phospho-dependent 14-3-3 binding was interpreted as evidence that phosphorylation promotes Bassoon exchange at synapses and increases active-zone scaffold dynamics.

6.6. Appendix 6: Modes of Synaptic Vesicle Endocytosis

Synapses employ multiple vesicle recycling mechanisms whose relative contribution depends on stimulation pattern and on the phosphorylation state of presynaptic endocytic proteins (Turner, Burgoyne and Morgan, 1999). Clathrin-mediated endocytosis (CME) is the classical retrieval pathway in which clathrin coats and adaptor proteins cluster vesicle proteins and deform the plasma membrane into coated pits that bud to regenerate synaptic vesicles, and it provides accurate cargo sorting during sustained activity (Murthy and De Camilli, 2003; Südhof, 2004). In addition, fast retrieval pathways operate at or near the active zone: in kiss-and-stay events, vesicles undergo brief fusion and remain close to the release site for rapid reuse, whereas in

kiss-and-run, a transient fusion pore opens and then closes without full collapse, allowing clathrin-independent local recycling (Südhof, 2004; Harata et al., 2006). Under intense stimulation, when exocytosis saturates these modes of retrieval, bulk endocytosis internalizes large plasma-membrane invaginations to form endosomal cisternae from which new vesicles are regenerated (Murthy and De Camilli, 2003). Phosphorylation of relevant components of the endocytic machinery biases synapses between these modes, thereby tuning presynaptic plasticity in an activity-dependent manner.